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SUPERSEDING
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February 16, 1955

FEDERAL SPECIFICATION

CALIPERS, GENERAL PURPOSE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers general purpose calipers used for measuring internal and external dimensions.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. The general purpose calipers covered by this specification shall be of the following types, classes, and styles, as specified (see 6.1):

Type I. Inside.
  Class 1. Spring.
    Style A. Rectangular leg.
    Style B. Round leg (tool makers).
  Class 2. Lock-joint.
  Class 3. Transfer.
  Class 4. Firm-joint.
  Class 5. Thread.
Type II. Outside.
  Class 1. Spring.
    Style A. Rectangular leg.
    Style B. Round leg (tool makers).
  Class 2. Lock-joint.
  Class 3. Transfer.
  Class 4. Firm-joint.
  Class 5. Thread.
Type III. Hermaphrodite.
Type IV. Slide.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specification and standard of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools

Federal Standard:

FED-STD-H28 - Screw-Thread Standards for Federal Services

(Activities outside the Federal Government may obtain copies of Federal Specifications, and Standards as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services Administration, Specifications Unit (WFSIS), 7th and D Street SW, Washington, DC 20407.)

FSC 5210

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 Materials. Unless otherwise specified, the material used in the manufacture of calipers shall be steel. The chemical composition and heat treatment employed shall conform to standard practices of manufacturers producing calipers of the types called for in this specification.

3.2 Design. The calipers shall be composed essentially of the components or parts specified herein for each type. To facilitate fabrication, it will be permissible to build up parts from separate elements, and in such cases, the various elements composing a part shall be rigidly held together, but in no case shall solder be used as a connecting medium.

3.2.1 Design of contact points. Contact points of legs shall be smoothly finished and ground similar to figure 1A for type I calipers and figure 6A for type II calipers. Type I calipers shall have a smooth radius on contact points.

3.3 Screw threads. Screw threads shall conform to FED-STD-H28.

3.4 Heat treatment. All parts of the tools shall be heat-treated in accordance with standard commercial practice in such a manner as to obtain the best overall characteristics of the steel used, and shall be stable after final heat treatment.

3.5 Type I, caliper, inside.

3.5.1 Class 1, spring.

3.5.1.1 Style A, rectangular leg.

3.5.1.1.1 Size of calipers. The calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in the sizes shown in table I.

TABLE I. Type I, class 1, styles A and B calipers.

Size	Distance across contact points in closed position (max)
Inches	Inch
3	3/8
4	1/2
6	5/8
8	11/16

3.5.1.1.2 Design. Calipers shall be similar to style A of figure 1 and shall consist essentially of a circular band spring, two legs, nut, washer, threaded steel rod, and swivel. All moving or sliding parts of the calipers shall be so designed and constructed as to permit smooth movement and operation.

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3.5.1.1.3 Circular band spring. The calipers shall be provided with a suitably hardened and tempered tool steel circular band spring at the hinged end which will cause the calipers to open, keep a sufficient tension on the caliper legs throughout their entire movement, and designed to remain permanently in position when in use.

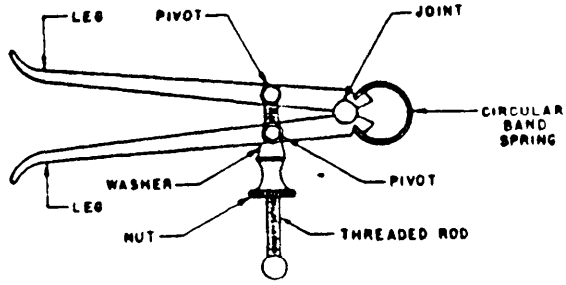
3.5.1.1.4 Legs. The legs shall be rectangular in the cross section and curved slightly outward at the ends for inside measurements. The ends of the legs shall be similar in shape and when in a closed position the maximum distance across the contact points shall be as shown in table I. When the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages, without making contact on the extreme sharp edges of the measuring ends.

3.5.1.1.5 Nut, threaded rod and swivel. The calipers shall be adjustable by means of a nut, either solid or quick adjusting as specified (see 6.1), travelling over a finely threaded rod, hinged or pivoted to one leg of the caliper and passing through an attachment fastened to the other leg and terminating in a small ball end to prevent the nut from coming off the threaded rod.

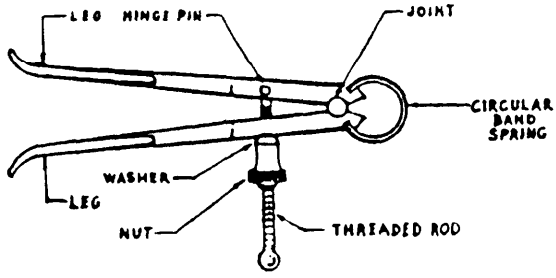
3.5.1.1.6 Washer. The nut of the calipers shall be fitted with a suitable washer or other means to facilitate its operation and reduce friction between the nut and the attachment on the leg of the caliper.

3.5.1.1.7 Finish. All exterior surfaces of the calipers, with the exception of threaded rods, knurled surfaces, and the insides of the circular band springs shall be polished. "Polished" shall mean a surface having a high degree of smoothness with the appearance of a good ground finish.

3.5.1.2 Style B, round leg (toolmaker's). The caliper shall be similar to style B of figure 1, and shall conform to the requirements specified in 3.5.1.1 except that the legs shall be round in the cross section, and the threaded rod shall pass through the body of one leg and be hinged to the other.



Style A



Style B

FIGURE 1. Type I, class 1, styles A and B calipers, inside, spring, rectangular leg and round leg.

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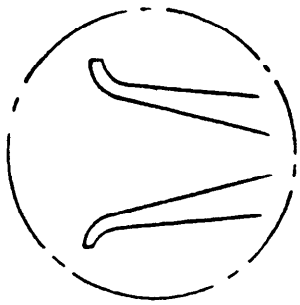


FIGURE 1A. Contact points, inside (Type I calipers).

3.5.2 Class 2, lock-joint.

3.5.2.1 Size of calipers. The calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in the sizes shown in table II.

TABLE II. Type I, class 2 calipers.

Size	Distance across contact points in closed position (max)
Inches	Inch
4	1/2
6	5/8
8	5/8
12	3/4
24	3/4

3.5.2.2 Design. The calipers shall be similar to figure 2 and shall consist essentially of two legs with a joint so constructed that the calipers may be locked in any desired position, and a sensitive adjustment device that will operate one leg of the calipers when the calipers are locked. All moving or sliding parts of the calipers shall be so designed and constructed as to permit smooth movement and operation.

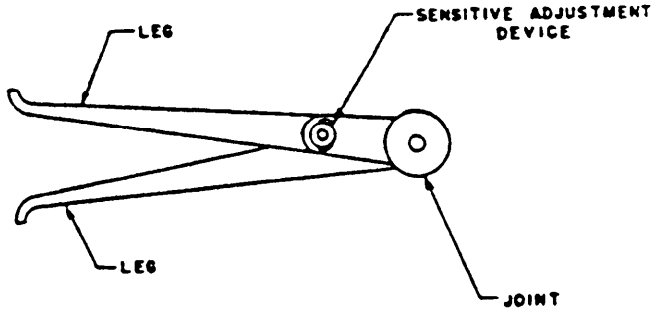


FIGURE 2. Type I, class 2, caliper, inside, lock-joint.

3.5.2.3 Sensitive adjustment device. The sensitive adjustment device shall be designed with a spring or other suitable means for taking up all backlash and assist the adjustable leg being held firmly at all settings of the calipers.

3.5.2.4 Legs. The legs shall be curved slightly outward at the ends for inside measurements. The ends of the legs shall be similar in shape and when in a closed position, the maximum distance across the contact points shall be as shown in table II. When the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under, and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

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3.5.2.5 Finish. All exterior surfaces of the calipers, with the exception of threaded rods and knurled surfaces shall be polished. "Polished" shall mean a surface having a high degree of smoothness with the appearance of a good ground finish.

3.5.3 Class 3, transfer.

3.5.3.1 Size of calipers. The calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in the sizes shown in table III.

TABLE III. Type I, class 3 calipers.

Size	Distance across contact points in closed position (max)
Inches	Inch
6	5/8
8	5/8
10	3/4
12	3/4

3.5.3.2 Design. The calipers shall be similar to figure 3 and shall conform to all requirements specified in 3.5.2, except that, in addition to the sensitive adjustment device, there shall be an auxiliary leaf on the other leg. This leaf shall remain in position but the leg shall swing free and return to position, showing the exact size measured.

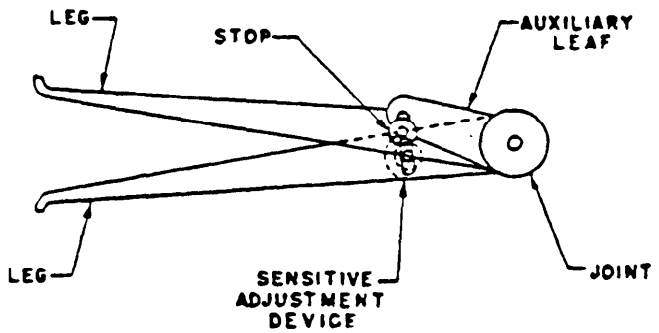


FIGURE 3. Type I, class 3, caliper, inside, transfer.

3.5.4 Class 4, firm-joint.

3.5.4.1 Size of calipers. The firm-joint calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the calipers. The calipers shall be furnished in the sizes shown in table IV.

TABLE IV. Type I, class 4 calipers.

Size	Distance across contact points in closed position (max)
Inches	Inch
6	5/8
8	5/8
10	13/16
12	7/8
18	1-3/16
24	1-7/16

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3.5.4.2 Design. The calipers shall be similar to figure 4 and shall consist essentially of two legs joined together at the pivot by means of a hexagonal or slotted head adjusting screw permitting the tension to be adjusted with a wrench or screwdriver. All moving or sliding parts of the calipers shall be so designed and constructed as to permit smooth movement and operation.

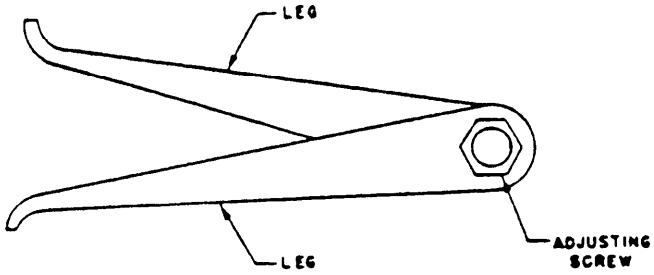


FIGURE 4. Type I, class 4, caliper, inside, firm-joint.

3.5.4.3 Legs. The legs shall be curved slightly outward at the ends for inside measurements. The ends of the legs shall be similar in shape and when in a closed position the maximum distance across the contact points shall be as shown in table IV. When the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

3.5.5 Class 5, thread.

3.5.5.1 Size of calipers. The calipers are designated in sizes which are defined as the distance between the center of the joints and ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 4-, 5-, and 6-inch sizes, as specified (see 6.1).

3.5.5.2 Design. The calipers shall be similar to figure 5 and shall comply with all the requirements specified in 3.5.1, except that the legs shall be as specified in 3.5.5.3.

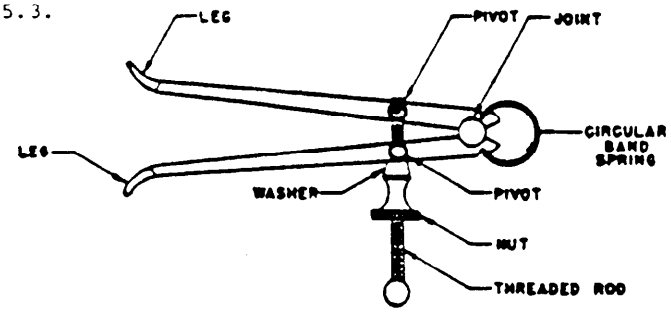


FIGURE 5. Type I, class 5, caliper, inside, thread.

3.5.5.3 Legs. The legs shall be curved slightly outward at the ends for inside measurements. The ends of the legs shall be similar in shape, and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch. The points of the legs shall be suitably shaped for measuring root and major diameters of screw threads. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

3.6. Type II, caliper, outside.

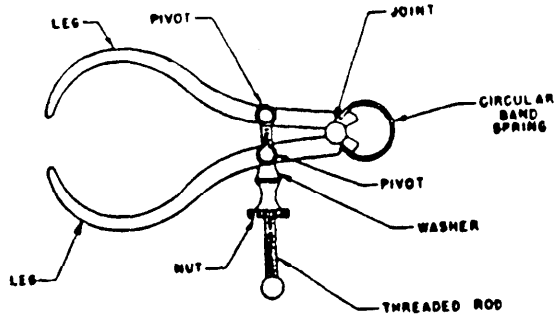
3.6.1 Class 1, spring.

3.6.1.1 Style A, rectangular leg.

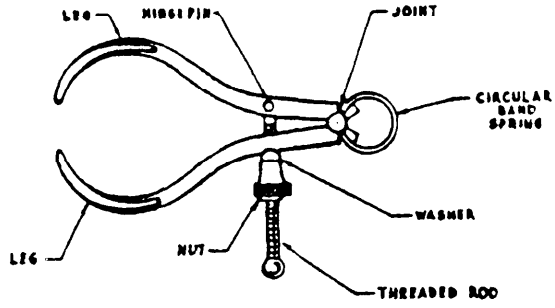
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3.6.1.1.1 Size of calipers. The calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 3-, 4-, 5-, 6-, 8-, 10-, and 12-inch sizes, as specified (see 6.1).

3.6.1.1.2 Design. The calipers shall be similar to style A of figure 6 and shall comply with all the requirements specified in 3.5.1, except that the legs shall be as specified in 3.6.1.1.3.



Style A



Style B

FIGURE 6. Type II, class 1, styles A and B, caliper, outside, spring.

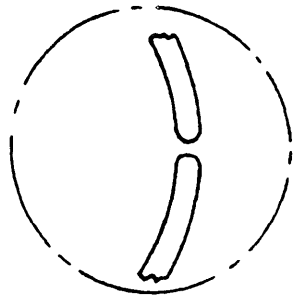


FIGURE 6A. Contact points outside (Type II calipers).

3.6.1.1.3 Legs. The legs shall be rectangular in the cross section and suitably bowed inwardly for outside measurements. The ends of the legs shall be similar in shape and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under, and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

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3.6.1.2 Style B, round leg (toolmaker's). The calipers shall be similar to style B of figure 6, and shall conform to the requirements specified in 3.6.1.1 except the legs shall be round in the cross section and the threaded rod shall pass through the body of the leg.

3.6.2. Class 2, lock-joint.

3.6.2.1 Size of calipers. The lock-joint calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 4-, 6-, 8-, 12-, and 24-inch sizes, as specified (see 6.1).

3.6.2.2 Design. The lock-joint calipers shall be similar to figure 7 and shall comply with all the requirements specified in 3.5.2, except that the legs shall be as specified in 3.6.2.3.

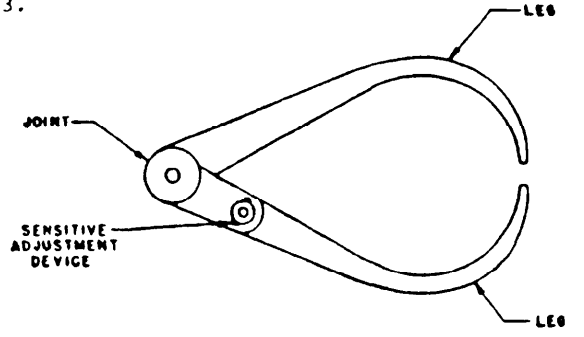


FIGURE 7. Type II, class 2, caliper, outside, lock-joint.

3.6.2.3 Legs. The legs shall be suitably bowed inwardly for outside measurements. The ends of the legs shall be similar in shape and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under, and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

3.6.3 Class 3, transfer.

3.6.3.1 Size of calipers. The transfer calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 8- or 12-inch sizes, as specified (see 6.1).

3.6.3.2 Design. The transfer calipers shall be similar to figure 8 and shall comply with all the requirements specified in 3.5.3, except that the legs shall be as specified in 3.6.3.3.

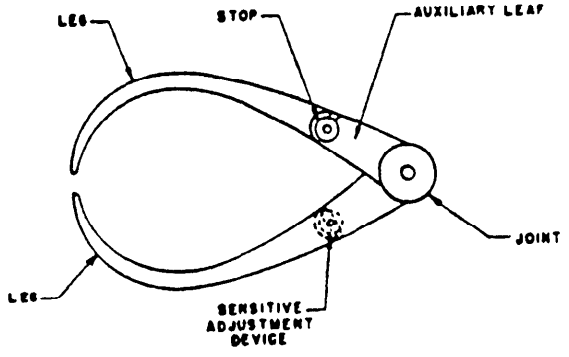


FIGURE 8. Type II, class 3, caliper, outside, transfer.

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3.6.3.3 Legs. The legs shall be suitably bowed inward for outside measurements. The ends of the legs shall be similar in shape and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under, and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages. The maximum error in transferring measurements shall not exceed 1/64 inch.

3.6.4 Class 4, firm-joint.

3.6.4.1 Size of calipers. The firm-joint calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 3-, 4-, 5-, 6-, 8-, 10-, 12-, 16-, 20- or 24-inch sizes, as specified (see 6.1).

3.6.4.2 Design. The firm-joint calipers shall be similar to figure 9 and shall consist essentially of two legs joined together at the joint by means of a hexagonal or slotted-head adjusting screw permitting the tension to be adjusted with a wrench or screwdriver. All moving or sliding parts of the calipers shall be so designed and constructed as to permit smooth movement and operation.

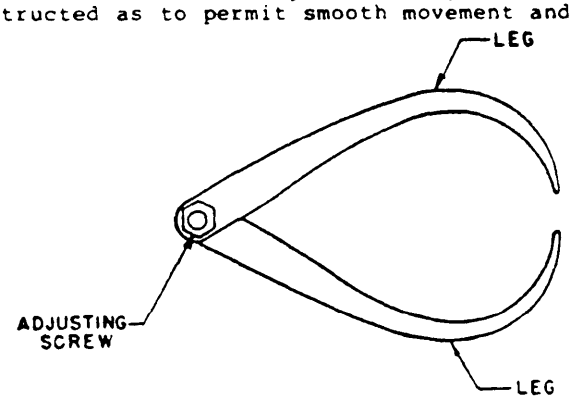


FIGURE 9. Type II, class 4, caliper, outside, firm-joint.

3.6.4.3 Legs. The legs shall be suitably bowed inwardly for outside measurements. The ends of the legs shall be similar in shape and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch for calipers 10 inches and under, and 1/32 inch for calipers 12 inches and over. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

3.6.5 Class 5, thread.

3.6.5.1 Size of calipers. The thread calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 4-, 5-, and 6-inch sizes, as specified (see 6.1).

3.6.5.2 Design. The thread calipers shall be similar to figure 10 and shall comply with all the requirements specified in 3.3.5, except that the legs shall be as specified in 3.6.5.3.

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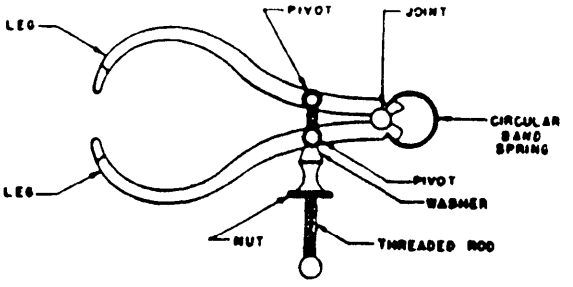


FIGURE 10. Type II, class 5, caliper, outside, thread.

3.6.5.3 Legs. The legs shall be suitably bowed inwardly for outside measurements. The ends of the legs shall be similar in shape and when the calipers are in a closed position, the contact points of the legs shall be equally distant from the joint within 1/64 inch measured from the joint and shall come evenly together in a lateral (flatwise) position within 1/64 inch. The points of the legs shall be suitably shaped for measuring root or minor diameters of screw threads. The measuring ends of the caliper legs shall be so rounded or shaped that all sizes within the capacity of the calipers may be readily transferred from measuring scales or gages.

3.7. Type III, caliper, hermaphrodite.

3.7.1 Size of calipers. Hermaphrodite calipers are designated in sizes which are defined as the distance between the center of the joints and the ends of the legs. The calipers shall open to a distance at least equal to the size of the caliper. The calipers shall be furnished in 4-, 6-, 8-, and 10-inch sizes, as specified (see 6.1).

3.7.2 Design. Hermaphrodite calipers shall be similar to figure 11 and shall consist essentially of two legs, joined together at the joint by means of a hexagonal or slotted-head adjusting screw permitting the tension to be adjusted with a wrench or screwdriver. All moving or sliding parts of the calipers shall be so designed and constructed as to permit smooth movement and operation.

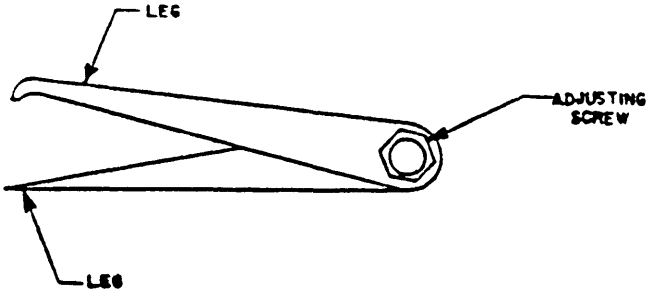


FIGURE 11. Type III, caliper, hermaphrodite.

3.7.3 Legs. One leg shall be curved slightly inward at the end; the other leg shall be similar to a divider leg. The divider leg shall be made of steel, suitably hardened and tempered, and shall terminate in a sharply shaped point at the end. The point shall be such as to avoid breakage, undue wear or bending under ordinary service conditions. The lateral (flatwise) misalignment of the measuring ends shall not exceed 1/32 inch.

3.7.4 Adjustable point. When specified (see 6.1), the divider leg shall have an adjustable point made of steel, suitably hardened and tempered, which may be adjusted and locked in position by a knurled thumbscrew. The leg shall be designed to permit the points (curved and adjustable) to come together and touch.

3.8. Type IV, caliper, slide.

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3.8.1 Size and capacity. The size of a slide caliper shall be defined as the overall length of the caliper. The capacity of a slide caliper shall be defined as being the span of obtainable inside and outside measurements between minimum and maximum on the slide. The calipers shall be furnished in the sizes and capacities shown in table V.

TABLE V. Type IV calipers.

Size (overall length)	Calipering capacity		Width of nibs on jaw	Depth of jaw	Thickness (approx.)
	Inside	Outside			
Inches	Inches	Inches	Inch	Inches	Inch
3	1/8 to 2-1/8	0 to 2	0.125	5/8 (min)	7/64
5	1/4 to 4	0 to 3-3/4	.250	1-7/16 ±1/16	1/8

3.8.2 Design. Slide calipers shall be similar to figure 12 and shall consist essentially of a frame with jaw, slide with jaw and stop, locking device, and with graduations on the frame and slide.

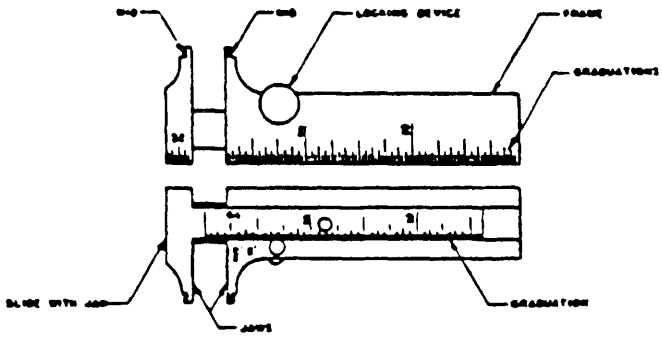


FIGURE 12. Type IV, caliper, slide.

3.8.3 Frame. The frame shall consist of a rectangular piece of substantial thickness having a fixed jaw as an integral part at one end and a longitudinal slot for guiding and retaining the slide. The measuring contact plane of the jaw shall be square with parallel sides of the longitudinal slot within 0.003 inch.

3.8.4 Slide with jaw. The slide shall consist of a rectangular piece terminating in a jaw as an integral part at one end. The measuring contact plane of the jaw shall be parallel with the measuring contact of the plane of the jaw on the frame. The rectangular portion of the slide shall run in the longitudinal slot of the frame. The slide shall have a stop so that it cannot be entirely withdrawn from the slot in the frame, but will allow movement of the slide to the capacity of the tool without excessive play when the locknut is completely loosened. A button on the face of the slide shall facilitate opening and closing of the calipers. The shape of the button shall be such that the thumb will not slide over the edge of the button when opening and closing caliper. When the locknut is loosened, it shall not require excessive force to move the slide in the frame of the caliper.

3.8.5 Jaws. The jaws shall have nibs designed to permit inside as well as outside measurements. The two nibs on each caliper shall be substantially equal in shape and size. The lateral (flatwise) misalignment of the jaws shall not be excessive when the jaws of the caliper are in any position from fully opened to fully closed and with the locknut loosened or tightened.

3.8.6 Locking device. The caliper shall be provided with a locking device having a knurled thumbscrew for locking the slide in any position.

3.8.7 Graduations. The sliding member shall be graduated in sixty-fourths of an inch for a distance of 2-1/2 inches for the 3-inch size, and for a distance of 4 inches for the 5-inch size, for reading outside and inside measurements Outside and inside measurements are read to lines, marked "out" and "in" on the frame, or a line and the plane formed by the face of the jaw. The back of the caliper shall be

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graduated in thirty-seconds of an inch along the full length of the caliper when the caliper is in a closed position. Graduations shall extend to the reading edge of the caliper. All graduations, reference lines, figures, and letters shall be clearly stamped, etched, or cut. In addition, all graduations shall be made to contrast with surrounding areas of the calipers by chemically etching the graduation lines or by other similar means so that graduations are distinct and easily readable. The maximum variation in width between like types of graduations and in the individual graduation shall be such that the maximum width shall not exceed the minimum width by more than 35 percent of the minimum width. The width of any graduation line shall be at least 25 percent less than the adjacent space width.

3.8.8 Finish. All surfaces and edges shall be true and smoothly ground, shall have a bright finish, and may be chrome plated.

3.8.9 Stainless steel. When specified in the contract or order (see 6.1), slide calipers shall be supplied in stainless steel.

3.8.10 Scale error. The scale error shall not exceed ± 0.003 inch between two graduations nonaccumulative.

3.8.11 Measurement error. The error in inside and outside measurements made with the caliper shall not exceed 0.006 inch when tested in accordance with paragraph 4.4.3.

3.9 Special features. When calipers with accuracy limits, scale graduations, or other features differing from those required by this specification are specified in the contract or order, the calipers shall conform to this specification insofar as applicable.

3.10. Finish. All finished exterior surfaces of calipers shall be such that surface roughness shall not exceed 63 microinches A.A.

3.11 Manufacturer's identification. Each caliper shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined, and the country of origin.

3.12 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of any tool that is otherwise in accordance with this specification.

3.13 Workmanship. The requirements within this specification are intended to describe the best commercial quality calipers available. The calipers shall conform to the quality of the end product specified by the requirements of this specification. In addition, the calipers shall be free from rust, burrs, fins, nodules, or other defects which may impair their serviceability, durability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection. Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.2.1 Inspection of the end item. Inspection of the calipers shall be performed as specified in the following paragraphs. Lot sizes shall be expressed in units of calipers for the purpose of determining sample sizes in accordance with MIL-STD-105. The sample unit shall be one completely fabricated caliper of one type, class, style, and size.

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4.2.2 Visual examination. Each sample caliper shall be examined for defects set forth below. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 defects expressed in defects per hundred units.

Examine	Defects
Finish	Not polished. Fins, burrs, sharp edges, or rough surfaces. Evidence of rust.
Design	Any characteristic not in accordance with specified requirements.
Construction and workmanship	Part missing. Fractures, bent, bowed, broken, malformed, or otherwise impaired. Misplaced, loose, or not in proper alignment. Functional component that requires abnormal force to operate. Any adjustable assembly that cannot be adjusted to perform the function intended.
Threads	Missing, broken, stripped, burred, or otherwise damaged.
Assembly	Not properly assembled or secured or does not function as required.
Marking identification	Manufacturer's name or trademark or country of origin missing, incomplete, not legible, not permanent.

4.2.2.1 Dimensional examination. Each sample caliper shall be examined dimensionally to determine compliance with the dimensional requirements specified herein. Any dimension not within specified tolerance shall be classified a defect. The inspection level shall be S-4 with an AQL of 2.5 defects expressed in defects per hundred units.

4.2.2.2 Testing of the end item. Each sample caliper shall be tested as specified in 4.4.1, 4.4.2, 4.4.3, and 4.4.4 to determine compliance with the requirements of 3.3, 3.8.10, 3.10, and table V (calipering capacity), respectively. The inspection level shall be S-4 with an AQL of 1.5 defects expressed in defects per hundred units.

4.3 Examination of preparation for delivery. Inspection to determine compliance with preparation for delivery shall be made in accordance with PPP-P-40.

4.4 Test procedures.

4.4.1 Surface roughness. Surface roughness shall be determined at representative spots by commercially recognized instruments for surface roughness measurement.

4.4.2 Threads. Threads shall be checked by means of thread plug and ring gages.

4.4.3 Slide calipers shall be checked against test gages of known dimensions. Tests shall be made near the minimum and maximum range of the calipers.

4.4.4 Scale accuracy. The graduated scales of the type IV calipers shall be tested for compliance with accuracy requirements of paragraph 3.8.10. These tests shall be made with suitable master rules, instruments, or laboratory apparatus of known accuracy. Tests shall be conducted at 68 degrees F., measurements at other temperatures shall be adjusted to values corresponding to the standard temperatures of 68 degrees F.

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4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing, and marking requirements comply with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style of caliper required (see 1.2.1).
(c) Size required (see applicable table and 3.5.5.1, 3.6.1.1.1, 3.6.2.1, 3.6.3.1, 3.6.4.1, 3.6.5.1, and 3.7.1).
(d) Solid or quick adjusting nut required (see 3.5.1.1.5).
(e) Adjustable point required for type III (see 3.7.4).
(f) Stainless steel required for type IV (see 3.8.9).
(g) Special features, if required (see 3.9).

MILITARY INTERESTS:

Custodian:

Air Force - 99

Review Activity:

Air Force - 84

PREPARING ACTIVITY:

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